Computational Identification of High-Altitude Molecules Using Molecular Networking and Liquid Chromatography High-Resolution Mass Spectrometry (LC-HRMS): Developing Software by Chemists, for Chemists

Independent Study Thesis

Presented in Partial Fulfillment of the Requirements for
the Degree Bachelor of Arts in the
Departments of Chemistry & Computer Science at The College of
Wooster

by
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Abstract

Include a short summary of your thesis, including any pertinent results. This section is *not* optional, and the reader should be able to learn the meat of your thesis by reading this (short) section.

This work is dedicated to the future generations of Wooster students.

Acknowledgments

No part of my journey at The College of Wooster has been accomplished alone. Behind every class, project, difficult conversation, and moment of growth are people who have supported me, challenged me, and reminded me why I started. This Independent Study is no exception. I am grateful to everyone who helped me become not only a better student and researcher, but also a better person.

First, I want to thank my Senior Independent Study advisors, Dr. Faust and Dr. Musgrave, for guiding me through a project that brings chemistry and computer science together. Dr. Faust, you have been one of my biggest champions. Thank you for believing in me, advocating for me, and helping me see what I am capable of even when I have struggled to see it myself. Dr. Musgrave, thank you for your guidance, patience, and willingness to work through the challenges of turning an interdisciplinary idea into a meaningful computer science project. I am also grateful to Dr. Paul Edmiston for bringing the original molecular-networking problem to my attention and giving me a reason to explore how the two disciplines I love could work together.

To the Computer Science Department, thank you for teaching me how to think critically, build things, and persist when the first solution fails. Dr. Heather Guarnera, thank you for always helping me find a way to make the impossible seem possible. Dr. Drew Guarnera, as my first computer science professor, you helped build a foundation that continues to shape my work. Dr. Palmer, thank you for contributing to my growth as a programmer and computer scientist. I am grateful to the rest of the department as well, including the faculty and staff whose time, encouragement, and teaching helped me find my place in the field.

To the Chemistry Department, thank you for making science something I wanted not only to study, but to spend my life exploring. Dr. Martin, thank you for your guidance as my academic advisor and for supporting my path across two majors. Dr. Snider, thank you for the difficult but necessary conversation that helped me grow. Dr. Bonvallet, thank you for making lectures something to look forward to. Dr. Gallo, thank you for helping me fall in love with chemistry. Dr. Baker, thank you for being one of my favorite professors and one of my favorite people to stop and say hello to. Dr. Feierabend, thank you for pushing me to be great, especially in the moments when I did not fully believe I could be. I am grateful to every other member of the Chemistry Department who shared their knowledge, offered encouragement, or made the department feel like a community.

To my best friends, Kate Carson, Eliza Davis, and Brooke Bradley, thank you for being part of my life through the busy days, the stressful weeks, and the moments worth celebrating. Your friendship has meant more to me than I can put into a single paragraph. Thank you for giving me people I could laugh with, rely on, and be myself around.

To my brothers of Beta Kappa Phi, thank you for giving me a brotherhood and a home within Wooster. I especially want to recognize Shea McLaren, Cooper Norwell, Gabriel Jones, Nathan Krieg, Parth, Isaac Prince, Jackson Ferrell and Matthew Dalton, my littles, Hayden Jacoby and Peter Dentzer, my grand big and big. Each of you has helped shape what brotherhood, responsibility, and friendship mean to me. I am proud of the memories we have made and the people we have become together. The first and finest will always hold a special place in my heart.

To the Residence Life Office, thank you for the opportunity to grow as a leader and learn what it means to show up for a community. I also want to thank Maddie Cuckow, my former AD and now Greek Life Director, for being part of that journey and for the support and guidance you have given me along the way.

I want to give a special thank-you to Lauren Richmond, who has been an important part of my life and personal growth throughout the years. Although life has taken us in different directions, I will always be grateful for the love, encouragement, and support she has given me. The memories we created, the challenges we faced, and the experiences we shared have shaped me in ways I will always appreciate. Her impact on my journey extends far beyond our time together, and I will always value the role she played in helping me become the person I am today.

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To Fr. David, thank you for walking alongside me during my journey toward baptism. Your guidance helped me through a deeply personal part of my life, and I am grateful for the time, care, and faith you shared with me.

Most importantly, I thank my family and siblings, especially my seven sisters and six brothers. You are part of the reason I care so deeply about using what I learn to make a difference. Thank you for the love, sacrifices, patience, and encouragement that have carried me through every stage of my education. Whatever I accomplish, I will always carry my family with me.

Finally, to everyone whose name may not appear here but who has offered a kind word, a difficult lesson, a new opportunity, or a reason to keep going, thank you. This work may have my name on its cover, but the person who completed it was shaped by all of you.

Vita

Raelyn M. Brooks is a candidate for the Bachelor of Arts degree in Chemistry and Computer Science at The College of Wooster, with an anticipated graduation date of May 2027. His academic interests lie at the intersection of computational chemistry, software engineering, and scientific research. He is particularly interested in developing computational tools that allow scientists to investigate complex chemical problems more efficiently and reliably.

During his undergraduate studies, Brooks pursued interdisciplinary research combining chemistry and computer science. His Senior Independent Study, advised by Dr. Jennifer Faust and Dr. John Musgrave, focuses on developing molecular-networking software to support the investigation of unknown molecular features in remote environmental samples using liquid chromatography–high- resolution mass spectrometry (LC-HRMS). His previous research includes developing a Python-based cheminformatics pipeline using RDKit to explore multistep organic synthesis pathways across large chemical reaction datasets.

Brooks has gained professional experience in software engineering and cloud computing. As a Software Developer Intern at the Federal Reserve Bank of St. Louis, he contributed to production software development, automated infrastructure, and security remediation. He has also worked as an Associate Cloud Engineer at GoHub LLC, developing cloud infrastructure and software for a transportation platform. Additionally, he has served as a coding instructor at iD Tech, teaching programming, software development, and problem-solving to hundreds of students.

Outside the classroom, Brooks has been actively involved in campus leadership and community engagement. He has served as President of Beta Kappa Phi, participated in Black & Code, and contributed to student life through Residence Life and other campus organizations. These experiences have strengthened his commitment to mentorship, leadership, and creating opportunities for others.

Following graduation, Brooks intends to pursue a career in software engineering, particularly in developing computational tools for chemistry and scientific research. He also hopes to pursue graduate study in chemistry, computer science, or an interdisciplinary field connecting the two disciplines.

Fields of Study Major field: Chemistry & Computer Science

Specialization: Computational Chemistry, Molecular Networking, and Liquid Chromatography–High-Resolution Mass Spectrometry (LC-HRMS)

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Preface

I have always considered myself a problem solver. Whether I am writing code, working through a chemical reaction, or trying to understand why something does not work as expected, I am drawn to problems that require patience, creativity, and a willingness to approach things differently. This is ultimately what attracted me to both chemistry and computer science. Chemistry has taught me to think beyond what I can immediately observe and to consider the countless interactions that explain the world around us. Computer science has taught me that failure is often part of progress, and that solving a problem sometimes requires starting over, questioning an assumption, or finding an entirely different approach. Together, these disciplines have shaped how I think, how I learn, and the kind of work I hope to pursue.

The inspiration for this Independent Study came from a conversation with Dr. Paul Edmiston, who introduced me to a challenge involving molecular-networking software. His students were exploring how molecular networking could help identify chemically modified substances relevant to forensic toxicology and support investigations involving drug-facilitated crimes. However, understanding the computational processes behind the software and evaluating the reliability of its results presented a challenge. Dr. Edmiston needed someone willing to look beneath the surface of the software and understand how it worked. As a brother to seven sisters, the opportunity to contribute to research with potential implications for personal safety and justice resonated with me deeply. Beyond its scientific importance, I recognized a chance to apply my knowledge of chemistry and computer science to a problem that could make a meaningful difference in people's lives.

What made this opportunity especially exciting was that it brought together the two fields I had spent my undergraduate career studying. I have never viewed chemistry and computer science as competing interests. Instead, I see them as disciplines that strengthen one another. Chemistry presents complicated questions about the world, while computer science provides tools for investigating those questions at a scale and level of detail that would otherwise be difficult to achieve. Although the research eventually developed into a project centered on molecular networking and the identification of unknown features in remote environmental samples, that initial conversation remained central to my motivation: I wanted to understand how the software worked and how it could be made more useful and reliable for the scientists who depend on it.

My experiences outside the classroom have further shaped that goal. During my internship as a software developer at the Federal Reserve Bank of St. Louis, I learned the importance of writing dependable software, testing my work, and understanding the consequences of decisions made within complex systems. Working with cloud infrastructure and software architecture has shown me how careful design affects reliability and long-term usefulness. As a coding instructor, I have taught hundreds of students how to approach difficult programming problems, often reminding them of the same lesson I continue to learn myself: not understanding something immediately does not mean it cannot be understood. My previous computational chemistry work, including developing tools to explore multistep organic synthesis using large reaction datasets, also helped me recognize how powerful software can become when it is built with a clear understanding of the science it supports.

After graduation, I hope to pursue a career developing software for chemists and other scientific researchers. I want to build tools that do more than produce results. I want researchers to understand those results, explore the evidence behind them, and have confidence in the computational processes they use. This Independent Study represents my first substantial attempt to bring that ambition into practice. It has also challenged me to recognize that being both a chemist and a computer scientist does not mean having every answer. Instead, it means being willing to learn from both disciplines, recognize the limitations of each, and do the work necessary to connect them.

I would not have reached this point without the faculty members who have challenged, supported, and believed in me throughout my time at The College of Wooster.

Within the Computer Science Department, I want to thank Dr. Heather Guarnera for consistently helping me find ways to make what once seemed impossible feel achievable. Your encouragement has pushed me to think bigger about what I am capable of accomplishing. I thank Dr. Drew Guarnera, my first computer science professor, for helping establish the foundation upon which so much of my work now rests. I am also grateful to Dr. Palmer for contributing to my growth as a computer scientist and to Dr. Musgrave, my Senior Independent Study advisor, for guiding me through the computational challenges of this interdisciplinary project.

Within the Chemistry Department, I owe a special thank-you to Dr. Faust, my Senior Independent Study advisor and one of my greatest champions. Your belief in my abilities, your encouragement, and your willingness to help me grow have meant more than I can adequately express. I thank Dr. Martin, my academic advisor, for helping guide my academic journey and supporting my interests across two demanding disciplines. Dr. Snider, thank you for the difficult but necessary conversation that helped me recognize where I needed to grow. Sometimes the advice we need most is not the advice we expect to hear. I thank Dr. Bonvallet for being a lecturer whose teaching has consistently made learning chemistry engaging. I thank Dr. Gallo for

helping me fall in love with chemistry and recognize just how much there is to discover within it. Dr. Baker, thank you for being one of my favorite professors and for the simple kindness of always being someone I look forward to greeting. Finally, Dr. Feierabend, thank you for continuing to challenge me to pursue excellence, particularly during moments when I have struggled to recognize my own potential.

I also want to thank Dr. Edmiston for presenting the problem that ultimately helped me discover the direction I want to take my career. Sometimes the most important opportunities begin with a conversation and a question that nobody has fully answered.

This Independent Study is more than a requirement for graduation. It represents the intersection of my interests, my experiences, and the people who have helped shape my education. More importantly, it is a beginning. I entered college wanting to understand how things work. I hope to leave having taken my first meaningful steps toward building things that help others understand the world.

CHAPTER 1

Introduction

1.1 Research Problem and Motivation

Anthropogenic chemicals enter the environment through industrial activities, agricultural practices, and the manufacture and use of consumer products. These inputs include pharmaceuticals, pesticides, industrial chemicals, and plastic-associated compounds, among other substances. Some chemicals are sufficiently persistent or mobile to move beyond their immediate sources, complicating efforts to determine their distribution and environmental fate [23, 4, 39]. Their occurrence is not confined to urban or industrial settings. Investigations of mountain regions and remote waters demonstrate that chemical pollution can be detected far from major emission centers, although the relative contributions of long-range transport and local activities differ among sites [9, 41, 22].

The analytical problem extends beyond determining whether a sample contains selected contaminants. Environmental samples may produce many signals in liquid chromatography–high-resolution mass spectrometry (LC-HRMS). These signals are processed into molecular features, which are characterized by measurements such as mass-to-charge ratio and retention time. However, detecting a feature does not establish the chemical identity of the substance responsible for it. Traditional targeted measurements are useful when the compounds of interest are known in advance; they are less suited to surveying chemical features that were not anticipated when a method was developed. Suspect screening and non-targeted analysis (NTA) broaden the range of features that can be investigated, yet incomplete reference libraries and uneven spectral coverage continue to constrain identification [23, 31, 17].

This limitation is especially important for remote environmental samples. A database match may support a candidate assignment for one feature, while other features in the same dataset remain unannotated despite containing potentially useful tandem mass spectrometry (MS/MS) information. Molecular networking offers a

way to examine these data by representing features as nodes and retaining relationships between features with sufficiently similar MS/MS spectra. The resulting network can help researchers explore candidate chemical relationships and prioritize features for further study [36, 27, 3]. However, a network connection is not itself a confirmed molecular identification [32].

The present Senior Independent Study addresses a second, connected problem: how to develop a dependable computational workflow that allows chemists to construct, inspect, preserve, and interpret molecular networks. Constructing these networks requires more than displaying nodes and edges. Software must correctly associate molecular features with their MS/MS spectra, calculate and filter spectral similarities using documented parameters, and preserve the underlying chemical metadata and computational settings. These requirements introduce challenges involving data structures, algorithms, software architecture, and reproducibility. For the College of Wooster's environmental HRMS research, this project therefore combines environmental chemistry with software engineering to develop and evaluate an interactive molecular-networking workflow. Environmental HRMS datasets provide both the motivating scientific application and the basis for chemical case studies.

1.2 Environmental and Chemical Background

1.2.1 Long-Range Transport of Organic Contaminants

Once released, an organic chemical's distribution depends on its physicochemical properties, its susceptibility to degradation, and the environmental processes acting on it. Atmospheric transport is one route by which certain pollutants can reach distant ecosystems. Volatilization, emissions associated with combustion or other activities, and the movement of particle-associated compounds can introduce chemicals into the atmosphere. During transport, reactions such as oxidation and photolysis may alter the original substances, producing transformation products with properties that differ from those of their precursors [9, 42]. The presence of a chemical in a remote sample, however, does not by itself establish its source or the pathway by which it arrived.

For some semivolatile organic compounds (SVOCs), repeated volatilization, atmospheric movement, and redeposition can contribute to long-distance redistribution. This sequence is often described as the **grasshopper effect** . Global fractionation and related temperature-dependent partitioning processes can favor the transfer of certain chemicals toward cooler regions, including mountain environments [38, 9]. These mechanisms are particularly relevant to persistent compounds whose transport and deposition can occur over

extended timescales. They should not be generalized to every pharmaceutical, pesticide, or other chemical detected by HRMS, because environmental behavior varies substantially among substances.

Mountain topography and climate influence where transported contaminants are deposited. Orographic precipitation can increase wet deposition as air masses rise and cool, while snowfall can scavenge some gas-phase and particle-associated chemicals. Temperature gradients also affect air–surface exchange and the partitioning of SVOCs among air, snow, vegetation, and other surfaces [5, 16, 10]. These processes operate together, rather than as independent explanations for every observation at a mountain site.

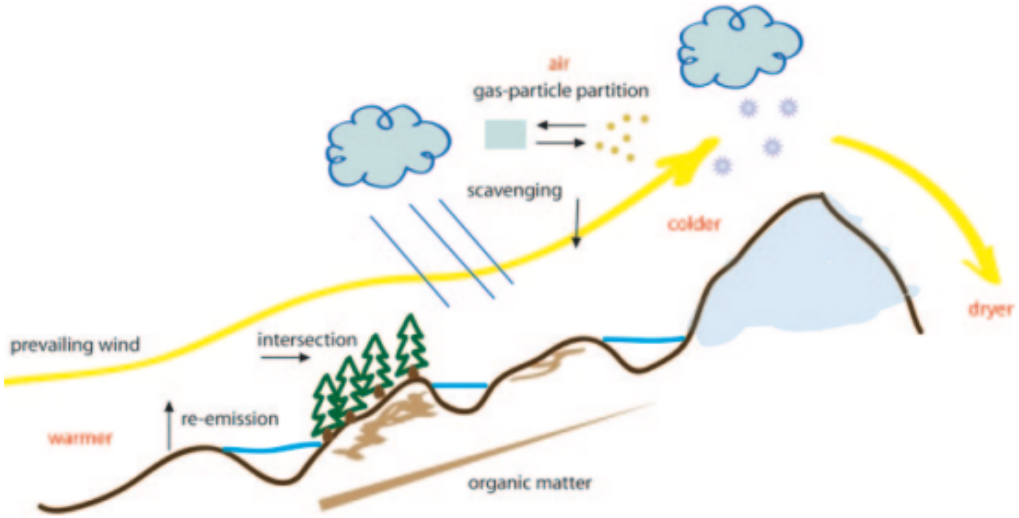


Figure 1.1: Conceptual pathways for atmospheric transport and deposition of contaminants in mountainous terrain, including precipitation scavenging, air–surface exchange, and elevation-dependent environmental conditions. Reproduced or adapted from the source discussed by Catalan [5]; confirm the original figure attribution.

1.2.2 High-Altitude and Remote Surface-Water Environments

High-altitude regions offer valuable settings for examining contaminant transport because their climatic conditions can favor deposition, seasonal storage, and subsequent redistribution of chemicals. Mountain snow and ice may temporarily retain atmospheric contaminants, and melting can transfer some of these substances into downstream waters. The timing and magnitude of release depend on snowpack processes and the physicochemical properties of the chemicals involved [16, 24]. These processes are important when interpreting surface-water samples because chemicals detected in water may reflect both recent inputs and substances previously deposited in snow or ice. However, detecting a chemical feature alone does not establish its ecological effects or the extent of exposure to organisms.

Research on mountainous environments has documented persistent organic contaminants and other human-associated chemicals in regions geographically distant from major industrial centers [9, 41, 37]. Studies of the Qinghai–Tibetan Plateau also demonstrate that contamination patterns can be influenced by human activity within a highland basin rather than solely by long-range atmospheric transport [22]. Consequently, remote environments should not automatically be considered free of local pollution sources. Tourism, settlements, transportation, and regional agricultural activities may contribute alongside atmospheric inputs, depending on the location and chemical involved. Distinguishing among these possible contributions requires additional environmental and analytical evidence.

The atmospheric transport and temperature-dependent partitioning mechanisms discussed in the preceding subsection provide a foundation for understanding how contaminants enter and move through mountain environments. Seasonal changes in temperature, snow cover, and precipitation can influence the balance between deposition, storage, and re-emission of semivolatile organic compounds [9, 16]. Figure 1.2 illustrates these processes along a mountain slope. These mechanisms help explain potential contaminant distributions, but they should not be interpreted as direct evidence of the transport pathways responsible for individual features detected in this project's environmental samples.

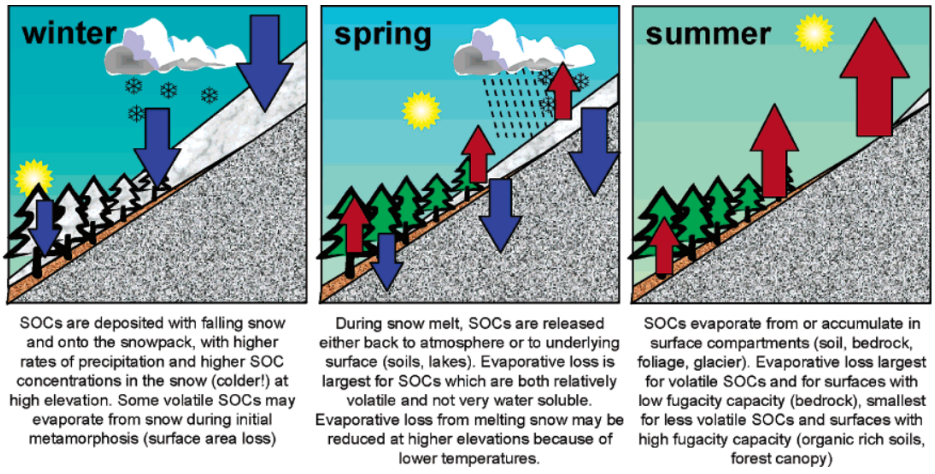


Figure 1.2: Illustration of seasonal air–surface exchange processes affecting semivolatile organic contaminants along a mountain slope. From the discussion of organic contaminants in mountains by Daly and Wania [9]; verify reproduction or adaptation permissions.

The broader mountain-contamination literature provides context for interpreting the environmental samples considered in this project. Daly and Wania reviewed investigations of persistent organic contaminants across mountain environments, emphasizing the influence of atmospheric transport, temperature, precipitation, and environmental retention [9]. Their compilation of earlier investigations, shown in Figure 1.3, illustrates the variety of mountain locations, chemical classes, and sampling approaches previously studied. This compilation

summarizes research activities rather than providing a standardized comparison of measured contaminant concentrations.

chemical	location (elevation[a])	date	ref
	OPPs and Other Currently Used Pesticides		
chlorpyrifos, diazinon, parathion	3 elevations (114/200, 533, 1920 m) in and below Sequoia National Park, Sierra Nevada, CA	15 days (Nov. 1990–Feb. 1991)	*26*
chlorpyrifos, methidathion		8 days (May–Oct. 1994)	*27*
chlorpyrifos, diazinon, malathion, chlorothalonil, trifluralin, endosulfan		5 days (May–Sept. 1996)	*28*
	SOCs		
PCBs in 2 HiVol samples	Lake Tahoe, Sierra Nevada, CA, USA (ca. 1900 m)	July 1995	*29*
OCPs in 7 HiVol samples	Bow Lake, Rocky Mtns., Canada (1940 m)	summer 1998	*61*
PAHs in 36 HiVol samples	Estany Redó, Pyrenees (2240 m)	1996–1997	*34*
	Gossenköllesee, Austrian Alps (2413 m)	1996–1997	
	Øvre Neådalsvatn, Norway (728 m)	1998	
PCBs and OCPs in 23 HiVol samples	Bow Lake (1974 m), Donald Station (770 m), Lower Kananaskis Lake (1667 m), Sundre, Rocky Mountains, Canada (1115 m)	June 1999–Aug. 2000	*33*
PCBs and OCPs in 20 HiVol samples	Pico de Teide, Canary Islands (47 and 2367 m)	June 1999–July 2000	*25*
α-HCH, HCB, DDT	Yahiko Mtn. (634 m), Japan	May–Nov. 2001	*30*
OCPs in 8 passive samples	Rocky Mountains, Canada (800–2300 m)	2000–2001 (annual average)	*31, 32*
PCBs and OCPs in 30 HiVol samples	Estany Redon, Central Pyrenees (2240 m) Skalnate Pleso, High Tatras (1778 m)	Nov. 2000–Jan. 2003	*106*
OCPs in 6 HiVol samples	Dingri (4340 m), Rongbu Valley (5050 m), Tibetan Plateau	May–June 2002	*16*

[a] Elevation given in meters above sea level.

Figure 1.3: Summary of selected mountain-air contaminant investigations, including reported chemical classes, study locations, and sampling periods, as compiled by Daly and Wania [9]. This is a study-summary table, not a comparison of measured concentrations. Verify the original figure attribution.

The present project is motivated by existing environmental HRMS datasets associated with remote and high-altitude locations, including the Himalayas, Mount Kilimanjaro, and the Qinghai–Tibetan Plateau. These environments provide an opportunity to investigate both previously recognized contaminants and chemical features that remain unannotated after reference-based screening. The exact samples, collection locations, analytical conditions, and data subsets used in the software evaluation will be documented in Chapter 2. Not every location discussed in the literature is necessarily represented in the datasets selected for the final case studies.

1.2.3 Previous Wooster Research and the Identification Gap

Previous research at the College of Wooster provides a direct foundation for this study. Fan investigated organic contaminants in snow and water samples from high-altitude locations, including Mount Everest, Mount Kilimanjaro, Qinghai Lake, the Niyang River, and several mountain snow sites. Using LC-HRMS-based suspect screening and semi-quantification, Fan reported contaminants across multiple chemical classes, including herbicides, pharmaceuticals, and human-associated compounds. Differences among sampling locations highlighted the importance of considering both potential long-range transport and local human

activity when interpreting contamination patterns [12]. Related research involving Wooster investigators has examined pesticides in rain and snow and applied suspect screening and non-targeted analysis to temperate snowfall [14, 26]. Together, these studies establish the analytical context for examining organic contaminants in complex environmental samples.

Fan's research also identified limitations that extend beyond the detection of individual contaminants. Sample preparation, ionization efficiency, and the selection of precursor ions for tandem mass spectrometry can influence which compounds produce usable analytical evidence. In data-dependent acquisition (DDA), for example, the instrument may preferentially select abundant ions for fragmentation, leaving some lower-abundance features without sufficient MS/MS information. Even when fragmentation spectra are available, incomplete reference libraries can prevent confident annotation [12]. These limitations demonstrate that the number of detected features does not necessarily reflect the number of compounds that can be reliably identified.

A related example of molecular networking comes from Coenen's undergraduate research at the College of Wooster. Although her work focused on designer benzodiazepines in forensic samples rather than environmental contaminants, she demonstrated how MS/MS spectral relationships could support the investigation of chemically related compounds, including an analogue not originally included in the reference network. Her research also discussed practical challenges associated with multistep GNPS workflows and the need for more accessible and reliable molecular-networking tools [7]. This work illustrates the broader applicability of spectral similarity analysis while identifying computational challenges relevant to the present project.

Together, these studies help define the identification gap addressed in this Senior Independent Study. From a chemistry perspective, researchers need methods for organizing analytical evidence and investigating features that remain unannotated after reference-based screening. From a computer science perspective, these methods require consistent spectral comparisons, appropriate data structures, traceable feature metadata, and software that allows researchers to inspect and preserve their analyses. The proposed Wooster molecular-networking workflow aims to support this process by connecting computational relationships to the underlying chemical evidence, without replacing reference-based identification or confirmatory analysis.

1.3 LC-HRMS, Suspect Screening, and Non-Targeted Analysis

1.3.1 LC-HRMS and Tandem Mass Spectrometry

Liquid chromatography–high-resolution mass spectrometry (LC-HRMS) combines chemical separation with accurate mass measurements to investigate complex mixtures. In liquid chromatography (LC), a liquid mobile

phase carries sample components through a column containing a stationary phase. Because compounds interact differently with these phases, they move through the column at different rates and elute at different times. The time at which a compound elutes is known as its retention time. This separation helps reduce interference among compounds in complex environmental samples, although some substances may still elute together. Following separation, compounds are converted into gas-phase ions through an ionization process such as electrospray ionization (ESI), allowing them to be detected by the mass spectrometer [23, 19].

High-resolution mass spectrometry measures the mass-to-charge ratio (m/z) of ions with high resolving power and mass accuracy. Resolving power describes the instrument's ability to distinguish ions with closely spaced m/z values, while mass accuracy describes how closely a measured value agrees with the expected value. During full-scan mass spectrometry (MS1), the instrument records ions over a selected mass range, providing information about the chemical components present in a sample. Computational processing of these measurements can identify molecular features characterized by properties such as m/z, retention time, signal intensity, and isotope patterns. Accurate mass measurements can help constrain possible elemental compositions, but they generally do not establish a unique molecular structure [19, 23].

Tandem mass spectrometry (MS/MS) provides additional chemical information by examining how selected ions fragment. An ion chosen for fragmentation is called a precursor ion. During collision-induced dissociation, the precursor ion collides with gas molecules and may break into smaller product ions. The resulting MS/MS spectrum records the m/z values and relative intensities of these fragments. Because molecular structures influence fragmentation behavior, these spectra can provide evidence about functional groups, molecular substructures, and possible chemical identities. Comparing experimental MS/MS spectra with reference spectra can support candidate annotations, while similarities between spectra can also help investigate relationships among features that lack direct library matches [19, 17].

The usefulness of LC-HRMS/MS therefore extends beyond detecting chemical signals. Combining chromatographic separation, accurate mass measurement, and fragmentation information provides multiple forms of evidence for investigating compounds in complex environmental mixtures. However, interpreting these measurements requires computational processing to detect features, associate them with the appropriate MS/MS spectra, and retain the analytical information needed for comparison. Software such as MS-DIAL supports these processing tasks, providing a connection between experimental measurements and the computational analyses developed later in this thesis [35]. The specific instrumentation, ionization modes, acquisition strategy, and processing parameters used for the environmental datasets are described in Chapter 2.

1.3.2 Molecular Features Versus Identified Compounds

A *molecular feature* is an analytical observation rather than a confirmed chemical identity. In LC-HRMS, features are generally characterized by properties such as mass-to-charge ratio (m/z), retention time, and signal intensity. However, a single compound can produce multiple observed signals through isotope patterns, adduct formation, or fragmentation. An adduct is an ion formed when a molecule becomes associated with another chemical species, such as a proton or sodium ion. In addition, different compounds can sometimes produce overlapping signals. For example, isomers have the same molecular formula but differ in the arrangement of their atoms, making them difficult to distinguish using accurate mass alone. Consequently, the number of detected features does not necessarily correspond to the number of unique chemical compounds in a sample. Likewise, a spectral library match or a relationship to an annotated feature does not automatically establish a confirmed chemical identity [23, 32].

Confidence in a chemical assignment must therefore reflect the available analytical evidence. Schymanski and colleagues proposed a five-level framework for communicating confidence in HRMS-based small-molecule identification, ranging from confirmed structures at Level 1 to exact-mass evidence at Level 5. Intermediate levels represent varying degrees of structural or molecular-formula information, depending on the available evidence [32]. Within this thesis, the term *feature* refers to a detected analytical signal without implying that its chemical identity has been established. A *candidate annotation* or *tentative identification* indicates that an assignment is supported by some analytical evidence but remains uncertain. Confirmed identification requires stronger evidence, such as appropriate comparison with an authentic reference standard. This distinction is particularly important in molecular networking, where connections between features may suggest chemical relationships without proving their molecular identities. The confidence associated with individual case-study features will be reported according to the supporting evidence rather than inferred from network appearance.

1.3.3 Suspect Screening and Non-Targeted Analysis

Environmental chemical analysis can be approached through targeted analysis, suspect screening, and non-targeted analysis (NTA). Targeted analysis focuses on a predefined group of compounds, often using established analytical methods and reference standards to support identification and quantification. For example, a targeted method might measure a selected group of pesticides known to occur in environmental waters. Suspect screening expands this approach by examining acquired data for compounds on a broader list of chemicals of interest, even when authentic reference standards are not available for every candidate. NTA takes a wider approach by detecting and investigating chemical features without requiring that all potential compounds be specified before analysis [23, 31].

These strategies are complementary rather than mutually exclusive. Suspect screening can prioritize chemicals associated with known environmental concerns, while NTA allows researchers to investigate additional features that were not anticipated during method development. Published investigations of pesticides in rain and snow, as well as broader organic chemical mixtures in snowfall, demonstrate the value of these approaches for environmental LC-HRMS research [14, 26]. However, expanding the number of detectable features also increases the challenge of interpreting the resulting data. Researchers must determine which signals represent meaningful chemical observations, evaluate possible identities using available evidence, and prioritize features that require additional investigation.

These approaches also contribute to broader efforts to characterize the *exposome*, which encompasses environmental and other exposures experienced over time [23]. Although suspect screening and NTA can provide information about portions of this chemical landscape, a single analytical dataset cannot characterize the entire exposome. The present project focuses more specifically on examining chemical features and their spectral relationships in existing high-altitude environmental datasets, rather than measuring total exposure or evaluating human health effects.

1.3.4 Limitations of Current Identification Approaches

Despite advances in high-resolution mass spectrometry, many detected features remain without confident chemical annotations. Reference databases may not contain a relevant compound or its experimental MS/MS spectrum, while differences in acquisition conditions can complicate comparisons with available references. Additionally, incomplete fragmentation data, overlapping signals, and similarities among chemical structures can make it difficult to distinguish competing candidate identities. Reviews of environmental NTA and computational structural annotation emphasize these limitations, as well as the importance of data quality, appropriate analytical evidence, and careful reporting of identification confidence [31, 17, 13].

These challenges create a gap between *detection* and *interpretation*. Increasing the number of observed features does not necessarily increase the number of confidently identified compounds. Computational approaches can help organize analytical evidence, identify promising spectral comparisons, and prioritize features for additional investigation. However, these methods must preserve uncertainty rather than treating every computational relationship as a confirmed chemical conclusion.

The limitations are not exclusively chemical. The computational workflow must also preserve the relationships between detected features, their associated MS/MS spectra, and the parameters used during data processing. If these connections are lost or handled inconsistently, useful chemical information may become difficult to recover, compare, or interpret. Reliable data organization, transparent algorithms, and

reproducible processing are therefore important for supporting chemical investigations. Molecular networking offers one approach for examining relationships among detected features, and its scientific and computational foundations are developed in the following section.

1.4 Molecular Networking and Theoretical Foundations

1.4.1 MS/MS Fragmentation and Spectral Similarity

Tandem mass spectrometry (MS/MS) provides information about molecular structure by recording the product ions generated when a selected precursor ion undergoes fragmentation. The resulting spectrum consists of fragment mass-to-charge (m/z) values and their corresponding intensities. Because the bonds and structural arrangements within a molecule influence how it fragments, compounds with related structures may produce some of the same product ions or exhibit similar fragmentation patterns. These relationships provide the chemical foundation for comparing MS/MS spectra and investigating features that lack direct reference-library matches [36, 3].

From a computational perspective, an MS/MS spectrum can be represented as a collection of numerical pairs, where each pair contains a fragment m/z value and its measured intensity. This representation allows a computer to compare spectra mathematically rather than relying exclusively on visual inspection. However, two spectra cannot be compared meaningfully by simply assuming that their peaks appear in the same order. The software must first determine which fragment peaks correspond to one another. This process, known as *peak matching* or *spectral alignment*, typically uses a mass tolerance to account for small differences between measured fragment masses. Tolerances may be expressed in daltons (Da) or parts per million (ppm), depending on the comparison method and instrument characteristics [3].

Spectral preprocessing can further influence the quality of these comparisons. Experimental spectra may contain noise, low-intensity peaks, or signals that contribute little useful structural information. Computational procedures may therefore filter selected peaks, normalize intensities, or apply other transformations before calculating similarity. Intensity normalization places spectra on a comparable numerical scale, while filtering can reduce the influence of unwanted signals. These operations must be applied consistently because different preprocessing choices can change which peaks are matched and how strongly two spectra are considered related [3, 8].

Although structurally related compounds may share fragments, the relationship between molecular structure and spectral similarity is not one-to-one. Different compounds can generate overlapping product ions, and the same compound can produce different relative fragment intensities under different collision

energies or instrument conditions. Some comparison methods account for possible structural differences by allowing fragment alignments shifted according to precursor-mass differences. Such approaches can reveal additional relationships beyond direct fragment matching, but they do not establish the chemical explanation for those differences [3, 30].

Molecular networking uses these computational comparisons to organize spectral relationships across many features simultaneously. Instead of investigating each spectrum independently, researchers can examine groups of features whose fragmentation patterns satisfy defined similarity criteria. An annotated feature may provide useful context for neighboring features without direct library matches. These relationships can help prioritize candidate structures or chemical families for further investigation, provided that the underlying fragmentation evidence and identification uncertainty remain accessible [36, 27, 11].

1.4.2 Cosine Similarity and Molecular-Network Construction

Once MS/MS spectra have been represented numerically and their fragment peaks have been aligned, a similarity function can quantify the agreement between them. One commonly used approach is *cosine similarity*, which measures the relationship between two vectors according to their orientation rather than their absolute magnitude. For spectral comparisons, the vectors describe fragment intensities using a common representation established through peak matching or alignment. The resulting similarity score summarizes how closely the represented fragmentation patterns agree under the chosen computational procedure [36, 3].

Let $\mathbf{x}$ and $\mathbf{y}$ represent two aligned spectral intensity vectors containing p positions. The intensity values at position i are denoted by x_i and y_i, respectively. Their cosine similarity is calculated as

$$\cos(\mathbf{x}, \mathbf{y}) = \frac{\sum_{i=1}^{p} x_i y_i}{\sqrt{\sum_{i=1}^{p} x_i^2}\,\sqrt{\sum_{i=1}^{p} y_i^2}}. \tag{1.1}$$

For nonnegative intensity vectors with nonzero magnitudes, cosine similarity ranges from zero to one. A value approaching one indicates stronger agreement between the represented spectra, while a value approaching zero indicates little agreement under the selected alignment. The numerator measures the combined contribution of corresponding intensities, while the denominator normalizes the result using the magnitudes of the two vectors. This normalization allows spectra with different overall signal intensities to be compared, although it does not eliminate differences caused by spectral quality, acquisition conditions, or preprocessing.

An important distinction is that spectral cosine similarity depends on how the corresponding intensity positions are established. Before applying Equation 1.1, a comparison algorithm must decide which fragment peaks match within the specified mass tolerance and how unmatched peaks are represented. Implementations

may also apply intensity weighting, spectral filtering, or minimum matched-peak requirements. Modified-cosine approaches can permit certain shifted fragment alignments based on precursor-mass differences, allowing comparisons that consider both shared fragments and potentially related fragmentation patterns. Consequently, two workflows using the term *cosine similarity* may produce different scores if their preprocessing and matching procedures differ [3, 8].

Calculating a similarity score does not automatically create a molecular-network connection. A network-construction algorithm applies additional criteria to determine whether two features should be connected. For example, a workflow may require a minimum cosine score and a minimum number of matched fragments before retaining an edge. Other procedures can further restrict connections by keeping only sufficiently high-ranking neighbors or limiting the size of connected groups. Established GNPS workflows use combinations of these filtering approaches to organize spectral relationships [36, 27]. Because the filters determine which edges remain, different parameter choices can produce different network structures from the same underlying spectra.

The number of possible comparisons also introduces an important computational challenge. If a dataset contains n spectra and every unique pair is compared once, the number of pairwise comparisons is

$$N_{\text{comparisons}} = \frac{n(n-1)}{2}. \tag{1.2}$$

This corresponds to quadratic, or $O(n^2)$, growth in the number of comparisons. For example, 1,000 spectra require 499,500 unique pairwise comparisons, while 10,000 spectra require 49,995,000 comparisons. The actual computational time also depends on the number of fragment peaks, the matching algorithm, and any preprocessing or candidate-selection methods. Nevertheless, this growth illustrates why computational efficiency becomes increasingly important when analyzing large LC-HRMS datasets. Strategies that reduce unnecessary comparisons or efficiently organize candidate relationships may therefore influence the scalability of a molecular-networking workflow.

Figure 1.4 summarizes the progression from analytical data processing to molecular-network construction. LC-HRMS data are first processed to detect features and obtain associated MS/MS spectra. Spectral comparison methods then quantify relationships between those features, and graph-construction rules determine which relationships are retained. The resulting network provides a computational representation that can be examined through visualization and graph-analysis methods.

These mathematical and algorithmic principles establish the foundation for the computational workflow developed in this Senior Independent Study. A reproducible implementation must specify its spectral representation, peak-matching procedure, similarity calculation, and edge-selection rules. The particular

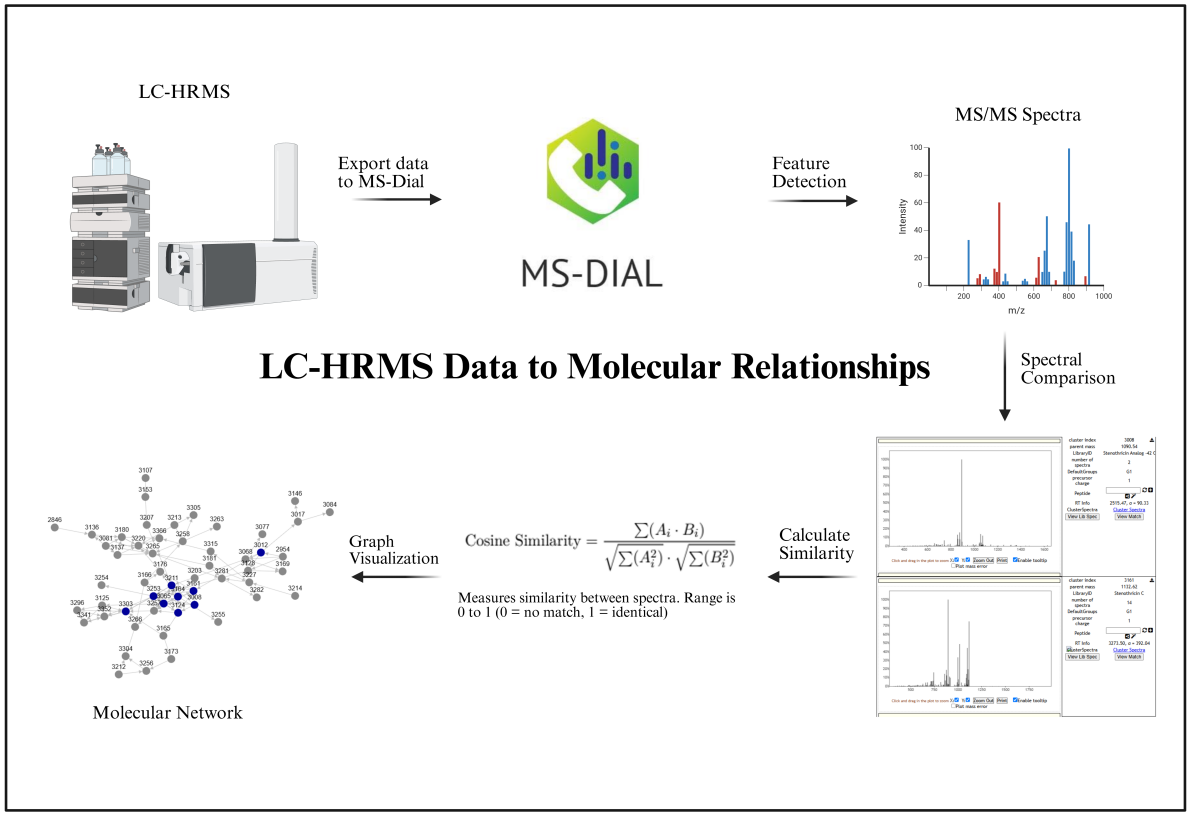


Figure 1.4: Conceptual workflow connecting LC-HRMS data processing to molecular-network construction. MS-DIAL supports feature detection and processing of mass spectrometry data. Associated MS/MS spectra can then be aligned and compared using spectral-similarity methods, including cosine similarity. Selected relationships are retained as edges in a molecular network, where nodes represent analytical features. Created with BioRender.com.

algorithms, parameters, and validation procedures used in the Wooster software will be documented in Chapter 2.

1.4.3 Graph Theory: Nodes, Edges, and Connected Components

Graph theory provides a mathematical framework for representing objects and the relationships between them. A graph is commonly written as $G = (V, E)$, where V is a set of vertices, or *nodes*, and E is a set of edges connecting selected pairs of vertices. In molecular networking, nodes can represent detected molecular features, while edges represent spectral relationships that satisfy the chosen computational criteria. Each node may also contain analytical information, such as precursor m/z, retention time, ionization polarity, or a candidate chemical annotation. Similarly, an edge may store a numerical similarity score and supporting information about how the relationship was calculated [25, 1, 36].

When the relationship between two features is based on a symmetric spectral-similarity calculation, the resulting network can be represented as an *undirected graph*. In an undirected graph, a connection between features u and v does not imply a direction from one feature to the other. A *weighted graph* additionally associates numerical values with its edges. For molecular networking, these weights may represent spectral similarity scores, although other relationship types require their own definitions. An edge weight describes the computed relationship under the selected method rather than the certainty of a chemical identification.

Computer programs must store graph relationships in structures that support efficient access and analysis. One common representation is an *adjacency list*, which maintains the neighboring vertices associated with each node. For example, if feature A is connected to features B and C, the adjacency representation records B and C as neighbors of A. An adjacency matrix provides another representation by storing relationships in a two-dimensional structure indexed by vertex pairs. These approaches involve different storage and access trade-offs. Adjacency lists are particularly useful for sparse graphs, where the number of retained edges is small relative to the number of possible connections. In contrast, an adjacency matrix stores a position for every possible vertex pair, requiring space proportional to the square of the number of vertices [25, 1].

Graph algorithms allow researchers to examine relationships that may be difficult to interpret from a network image alone. A *connected component* is a maximal group of vertices in which every pair is connected by some path through the graph. An *isolated vertex* has no retained edges, while the *degree* of a vertex is the number of edges incident to it in an undirected graph. These measures help describe network organization, identify disconnected features, and locate groups that may be useful for further chemical investigation. However, a connected component is not automatically a chemical family. Two features can belong to the same component through a sequence of intermediate connections without being directly similar to each other.

Connected components can be identified through graph-traversal algorithms such as depth-first search (DFS) or breadth-first search (BFS). Both approaches systematically explore the vertices reachable from a starting node. DFS follows a sequence of connections before returning to explore alternatives, while BFS explores neighboring vertices in successive layers. When implemented using an adjacency-list representation, a complete traversal can run in $O(|V| + |E|)$ time, where $|V|$ is the number of vertices and $|E|$ is the number of retained edges. This efficiency is useful because the resulting graph may contain substantially fewer edges than the number of spectral pairs initially considered.

Graph-analysis libraries provide reusable implementations of these data structures and algorithms. NetworkX, an open-source Python library, supports attributed graphs, weighted edges, connected-component analysis, graph traversal, and several graph-export formats [15]. Such functionality makes it a potential component of the Wooster molecular-networking software. However, NetworkX is a general-purpose graph library rather than a mass-spectrometry analysis system. It does not independently determine which fragment peaks match or whether two spectra are chemically similar. Those relationships must be calculated and validated through separate spectral-processing procedures before they can be represented as graph edges.

Figure 1.5 illustrates a conceptual molecular network with multiple connected features. The visual representation makes it possible to observe individual connections and groups within the network. Nevertheless, the scientific meaning of those relationships depends on the spectra, annotations, and computational criteria used to generate them.

These graph-theoretical concepts provide the mathematical basis for organizing spectral relationships within the proposed software. They also establish an important separation between chemical evidence and computational representation: spectral comparisons determine which relationships are supported, while graph data structures and algorithms make those relationships possible to store, analyze, and explore.

1.4.4 Network Visualization and Interpretation

Network visualization provides a way to examine relationships that may be difficult to recognize from tables of spectral similarities alone. Once a molecular network has been constructed, a visualization system assigns positions and visual properties to its nodes and edges. Features can be represented as circles or other symbols, while connecting lines indicate retained relationships. Color, node size, edge thickness, and labels can communicate additional information, such as annotation status, feature intensity, or spectral similarity. However, these visual properties must have clearly defined meanings so that the interface does not communicate more chemical certainty than the underlying evidence supports.

Graph-layout algorithms determine how vertices are positioned within a visualization. A commonly used

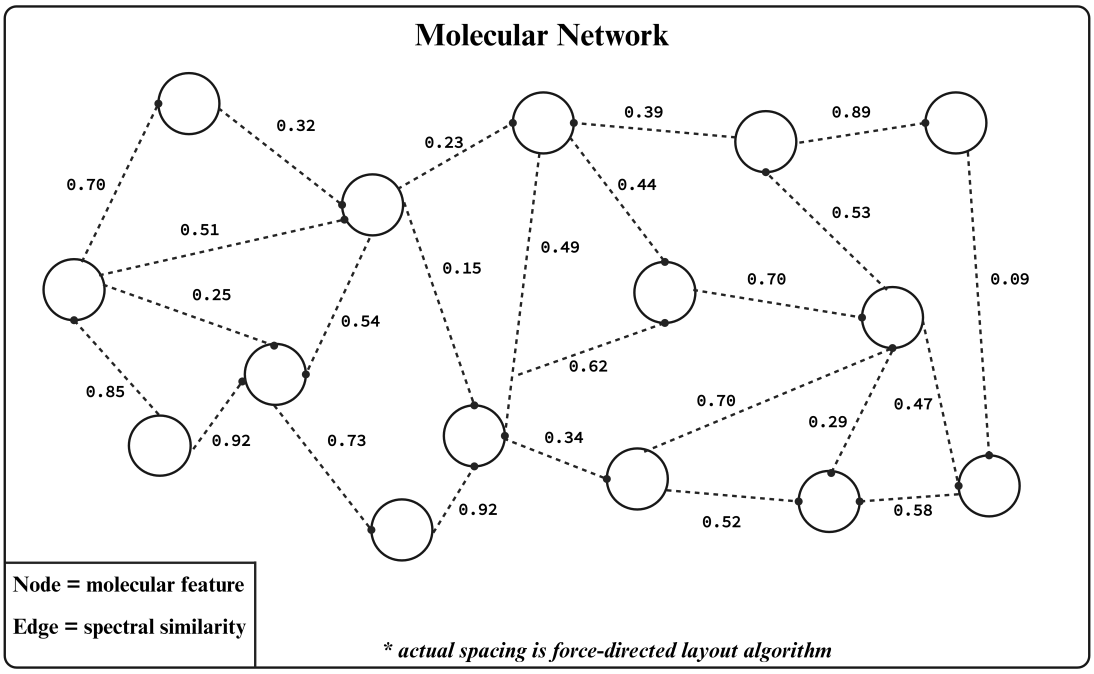


Figure 1.5: Conceptual example of a molecular network represented as a graph. Nodes correspond to detected molecular features, and edges represent retained spectral-similarity relationships. Connected groups illustrate graph organization but do not independently establish chemical identity or structural relationships. Created with BioRender.com.

approach is a *force-directed layout*, which models graph elements using simulated attractive and repulsive forces to produce a readable arrangement. Connected vertices tend to be drawn together, while repulsive forces help reduce overlap. ForceAtlas2 is one example of a force-directed layout method designed for network visualization [18]. These algorithms can make connected groups easier to recognize, but the resulting positions are mathematical display choices rather than experimental measurements. In particular, the physical distance between two nodes on a screen should not automatically be interpreted as a quantitative measure of chemical similarity.

Interactive visualization extends the usefulness of a network by allowing researchers to investigate individual features and relationships. Rather than viewing a static diagram, a user can select a node, examine its neighbors, inspect an associated MS/MS spectrum, or retrieve available information about its precursor mass, retention time, and candidate annotations. Selecting an edge can similarly provide access to its similarity score, matched-fragment evidence, and computational settings. Research on interactive graph exploration emphasizes the importance of connecting visual overviews with operations that support users' analytical tasks [29].

From a software engineering perspective, this functionality requires coordination between several components. A computational layer stores the graph structure and associated metadata, a visualization layer converts that information into graphical elements, and an interaction layer responds to researcher actions. These responsibilities can be separated so that changing the visual appearance of a network does not unintentionally change the underlying scientific calculations. For example, hiding weak edges in a browser may change the displayed network without recalculating spectral similarity. The software must distinguish such visual filtering from generating a new graph under different computational thresholds.

Another important consideration is *data provenance*, which refers to information describing the origin of data and the processes used to produce them. In molecular networking, provenance can include the original feature identifiers, associated spectra, software versions, preprocessing settings, similarity algorithms, and edge-selection parameters. Preserving this information allows researchers to understand why a connection exists and, where the necessary inputs are available, attempt to reproduce the analysis. Saving only the graphical positions of nodes and edges is insufficient because a network image does not contain all the analytical evidence or computational decisions used to construct it.

These principles are relevant to the chemist-centered software developed in this Senior Independent Study. A scientifically useful interface should connect the overall network view to the evidence supporting individual features and relationships. Researchers should be able to move between graph-level patterns and detailed chemical information without losing the context of the original analysis. The intended purpose is not simply to produce more attractive molecular networks, but to support transparent investigation, consistent

interpretation, and reproducible computational workflows. The specific website architecture, interactions, and evaluation methods will be described in later chapters.

1.4.5 Limitations of Spectral Similarity as Evidence of Chemical Identity

Spectral similarity provides useful evidence for investigating relationships among molecular features, but it cannot independently confirm the identity of an unknown compound. Different structures may produce shared fragment ions, while structurally related compounds can generate different spectra depending on ionization conditions, collision energy, and other experimental factors. Isomeric compounds may also exhibit closely related fragmentation patterns, making confident structural discrimination difficult without additional evidence. Consequently, a high similarity score supports agreement between spectra under a particular comparison procedure rather than proving that two features represent the same molecule or a confirmed structural analogue [3, 17].

Computational processing introduces additional sources of uncertainty. Peak filtering, intensity normalization, fragment-matching tolerances, and the selected similarity function can all influence the resulting scores. Network-construction thresholds and filtering rules then determine which relationships are retained. A stricter threshold may remove weaker connections, while a more permissive threshold may produce a denser graph containing additional uncertain relationships. These choices can alter connected components and the apparent organization of a network without changing the original experimental measurements [36, 27, 8].

Data integrity is another essential limitation. A molecular-networking algorithm depends on correctly associating each feature with its corresponding MS/MS spectrum and analytical metadata. Incorrect identifiers, missing spectra, or inconsistently processed data can generate misleading relationships or prevent meaningful comparisons. Furthermore, the absence of an edge does not establish that two compounds are chemically unrelated. It may instead reflect insufficient fragmentation evidence, low spectral quality, or the exclusion of a potential relationship by the selected computational criteria.

Chemical interpretation also requires care when examining precursor-mass differences or groups of connected features. A mass difference between two nodes may be consistent with a possible chemical modification, but it does not establish that one compound is a transformation product of another. Likewise, membership in a connected component does not prove a common environmental source or reaction pathway. These interpretations require supporting chemical evidence, and candidate annotations should retain confidence levels appropriate to the available information [32, 11, 6].

These limitations emphasize the difference between computational reproducibility and chemical identification confidence. A workflow may consistently reproduce the same similarity scores and network structure

while still containing uncertain chemical assignments. Conversely, useful experimental evidence may be difficult to interpret when the computational workflow does not preserve its source data and processing decisions. Both problems must therefore be addressed when developing software for chemical research.

The molecular-networking workflow proposed in this Senior Independent Study is intended to support evidence-based investigation rather than automatic structural confirmation. By preserving spectral information, documenting computational parameters, and making individual relationships accessible for examination, the software aims to help chemists prioritize unknown features while maintaining appropriate scientific uncertainty. This combination of chemical interpretation and transparent computation provides the foundation for the software requirements and research objectives discussed in the following sections.

1.5 Existing Software and Research Gap

1.5.1 MS-DIAL and GNPS in the Existing Workflow

Interpreting LC-HRMS/MS data requires multiple computational stages, each addressing a different analytical or mathematical problem. Raw mass spectrometry measurements must first be processed into identifiable features and associated fragmentation spectra. These observations can then be compared to investigate possible chemical relationships, and the resulting information must be organized into a form that researchers can examine. At the College of Wooster, MS-DIAL and the Global Natural Products Social Molecular Networking platform (GNPS) have provided important parts of this process. Although both programs support chemical analysis, they differ in their underlying responsibilities, computational methods, and approaches to presenting results [35, 36, 27].

MS-DIAL, which stands for Mass Spectrometry–Data Independent Analysis, is an open-source application developed for processing mass spectrometry data. Despite its name, the software supports both data-independent acquisition (DIA) and data-dependent acquisition (DDA), among other analytical methods. The program was originally developed to address complex MS/MS data processing and has expanded to support feature detection, chromatographic alignment, chemical annotation, and other analytical operations [35, 34, 33]. MS-DIAL 5 provides these capabilities through a desktop application built within the .NET software ecosystem, with separate computational and graphical components. This architecture allows established analytical processing methods to operate within an interface designed for mass spectrometry researchers.

The MS-DIAL processing workflow begins with the import of experimental mass spectrometry data. Its feature-detection procedures examine analytical signals to locate chromatographic peaks and characterize them using measurements such as mass-to-charge ratio (m/z), retention time, and signal intensity. When

multiple samples or analytical runs are processed, feature alignment attempts to associate corresponding observations across datasets using criteria such as mass and retention-time agreement. Available MS/MS spectra can then be associated with detected features, allowing reference-based annotation and downstream spectral analysis. These procedures help transform large collections of instrumental measurements into organized analytical records that researchers can interpret [35, 34].

An important aspect of MS-DIAL is its ability to export processed information for use in other software. MS-DIAL 5 documentation describes exports containing peak lists, alignment results, representative MS/MS spectra, and processing parameters. Spectral information can be exported in formats such as Mascot Generic Format (MGF) and Mass Spectral Library (MSP) format, while feature-level measurements can be represented in tabular files. MS-DIAL 5 also includes molecular-spectrum-networking functionality, meaning that network construction is not exclusive to GNPS. However, the availability of these capabilities does not automatically establish that every exported feature, spectrum, annotation, and parameter can be transferred into another application without additional validation. The particular export structures used in this project must be examined to determine how their records correspond.

GNPS provides a complementary computational environment for mass spectrometry analysis. Developed as a community-based platform, GNPS supports spectral-library searching, molecular networking, data sharing, and collaborative interpretation of MS/MS information [36]. Unlike software that primarily processes raw chromatographic data, GNPS places substantial emphasis on comparing fragmentation spectra and organizing their relationships. One established approach is classical molecular networking, in which submitted MS/MS spectra are organized through spectral comparisons and may be grouped into consensus representations. A second approach, feature-based molecular networking (FBMN), incorporates chromatographic features identified by upstream software and maintains relationships between those features and their associated spectra [27].

Feature-based molecular networking is especially relevant to the Wooster research workflow because it connects existing LC-HRMS processing with spectral-network analysis. In a typical MS-DIAL-to-GNPS workflow, MS-DIAL first produces a feature quantification table and a corresponding file of representative MS/MS spectra. The quantification table records feature-level information and measurements across samples, while the spectral file contains fragmentation information associated with those features. GNPS then uses these inputs to compare spectra, calculate similarity scores, apply networking criteria, and generate relationships that can be explored alongside available annotations and sample information [27]. Correctly matching feature identifiers between these inputs is essential because a network connection is scientifically useful only when it can be traced to the appropriate analytical observations.

Constructing a GNPS molecular network involves more than calculating a single similarity score. Its

workflows can apply fragment-mass tolerances, minimum matched-peak requirements, spectral-similarity thresholds, and additional filtering rules that determine which relationships remain in the graph. These decisions influence network density, connected components, and the relationships available for chemical interpretation. GNPS also provides access to reference spectral libraries that can support candidate annotations, although a library match does not independently guarantee confirmed chemical identity [36, 27]. Because these operations depend on specific computational settings, interpreting or reproducing a GNPS network requires more information than its final graphical appearance.

GNPS2 represents a newer, separately documented platform with dedicated workflows for feature-based molecular networking and other mass spectrometry analyses. Its FBMN workflow supports inputs such as feature quantification tables and representative MGF or MSP spectral files, together with optional sample metadata. GNPS2 therefore remains an important existing alternative when evaluating the need for a locally developed application. However, GNPS and GNPS2 should not be assumed to use identical interfaces, parameter defaults, or computational behavior. Meaningful comparisons require identifying the particular workflow, software version, input files, and settings used for each analysis.

Figure 1.6 illustrates the established MS-DIAL-to-GNPS workflow that provides a foundation for the present study. The figure emphasizes the movement from processed LC-HRMS/MS data to molecular-network relationships rather than depicting the final architecture of the proposed Wooster website.

A further computational consideration is how the resulting molecular network should be represented and analyzed within a new software system. NetworkX is an open-source Python library that provides data structures and algorithms for constructing, modifying, and investigating graphs [15]. It supports representations such as undirected graphs, weighted edges, and nodes containing additional attributes. Within a molecular-networking application, these capabilities could be used to store feature identifiers, spectral-similarity scores, chemical metadata, and graph relationships. NetworkX also supplies algorithms for examining neighbors, connected components, and other structural properties. However, it does not independently process LC-HRMS measurements, align MS/MS fragments, or determine whether two spectra are chemically related. Those operations require separate, scientifically appropriate computational procedures.

The relationship among these technologies helps define the potential architecture of the proposed system. MS-DIAL can remain responsible for established upstream analytical processing, while existing GNPS workflows provide scientific methods and results against which a local implementation can be evaluated. NetworkX offers a possible graph-processing component after spectral relationships have been calculated. The Wooster website would then connect these computational operations to an interface designed for local researchers. This division of responsibilities is a proposed architectural direction rather than a finalized

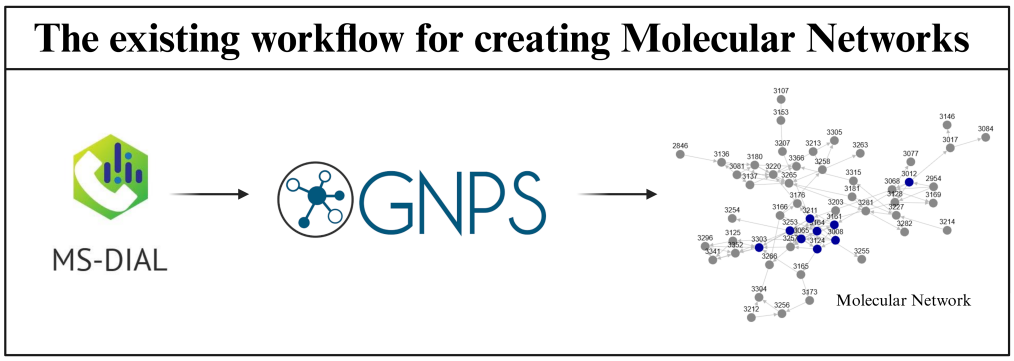


Figure 1.6: Overview of the existing molecular-networking workflow used in environmental HRMS research at the College of Wooster. MS-DIAL processes mass spectrometry data to obtain molecular features and associated MS/MS spectra, which can subsequently be analyzed through GNPS to investigate spectral relationships. This existing workflow provides a foundation for evaluating computational requirements and developing a Wooster-specific molecular-networking application. Created with BioRender.com.

implementation. The exact software components, import procedures, and algorithms selected for the project will be described and evaluated in Chapter 2.

1.5.2 Documented Limitations in the Wooster Research Workflow

The motivation for developing a Wooster-specific molecular-networking application emerged partly from practical difficulties reported during previous local research. These concerns included challenges involving the preservation, retrieval, and export of molecular-networking analyses, as well as the ability to investigate results efficiently. Coenen's undergraduate research provides relevant local context because it applied molecular networking to designer benzodiazepines and discussed practical challenges associated with the analysis workflow [7]. Although that research involved forensic rather than environmental samples, its computational concerns are relevant to the current project. These experiences identify laboratory needs that warrant investigation, but they do not establish that GNPS or other existing platforms are universally unreliable.

It is also important to distinguish operational concerns from the capabilities of the underlying software. MS-DIAL 5 and GNPS2 already provide substantial data-processing and molecular-networking functionality. The existence of newer software versions does not eliminate the need to evaluate how well those tools support the specific tasks performed by Wooster researchers. A workflow may be scientifically capable of generating a network while still requiring researchers to move among multiple programs, manage several output files, or reconstruct the settings used for a previous analysis. Such challenges concern how computational tools are integrated into the laboratory's research practices rather than whether the individual programs can perform their intended calculations.

One important technical challenge occurs when analytical information moves between software systems. A processed LC-HRMS feature may be represented by an identifier in an alignment table, while its associated MS/MS spectrum appears in another file. If these identifiers are inconsistent, missing, or incorrectly interpreted, an imported network may connect the wrong analytical records or omit information needed for chemical interpretation. Differences in file formats, retention-time units, intensity definitions, and missing-value conventions can introduce similar problems. These risks are particularly relevant when integrating software developed independently, because compatible file extensions do not necessarily guarantee compatible data structures or field meanings.

From a computer science perspective, these problems involve *data integrity*, which concerns the accuracy, consistency, and preservation of information as it is processed and transferred. For molecular networking, data integrity requires maintaining valid relationships among feature identifiers, associated spectra, candidate annotations, and their source records. A scientifically meaningful interface should not silently discard

unmatched records or present a network relationship without sufficient information about its origin. Instead, the computational workflow should define how inputs are checked, how invalid or incomplete information is handled, and how the resulting graph can be traced back to its source data.

A related requirement is *reproducibility*, or the ability to repeat a computational analysis using documented inputs, processing methods, and settings. A molecular network can change when fragment-matching tolerances, similarity thresholds, or filtering rules change, even when the underlying experimental data remain unchanged. Preserving only the final visualization is therefore insufficient. Researchers also need access to the input files, relevant software versions, parameter values, and outputs used to construct the network. Established scientific-computing practices emphasize organized data, documented computational procedures, version control, and repeatable analyses [40].

These concerns suggest that software quality should be evaluated through several related properties. *Correctness* concerns whether the implemented calculations and data relationships behave as specified. *Reliability* concerns whether the application performs its expected operations consistently and handles failures appropriately. *Reproducibility* concerns whether analyses can be reconstructed under documented conditions. Finally, *usability* concerns whether researchers can perform the necessary analytical tasks and understand the information presented to them. These properties are related but distinct: an application can calculate correct similarity scores while still being difficult to use, and a visually accessible network can remain scientifically incomplete if its supporting evidence cannot be retrieved.

For the present study, these concerns will be translated into explicit requirements and evaluation procedures rather than treated as problems already solved by the proposed software. Relevant tests include verifying feature-to-spectrum associations, checking the preservation of chemical metadata, comparing graph results under controlled settings, and determining whether saved analyses can be reopened and exported. Reported local difficulties will be distinguished from limitations reproduced through testing. This approach allows the research to evaluate the new application according to observable behavior rather than assumptions about existing software.

1.5.3 The Need for Software Designed Around Chemists' Needs

The availability of established software does not necessarily remove the need for applications designed around a particular research community. MS-DIAL, GNPS, and GNPS2 provide extensive analytical and computational capabilities, but their interfaces and workflows are intended to support a broad range of researchers and experimental applications. The requirements of the College of Wooster's environmental HRMS research may differ from those of other laboratories in how data are organized, how chemical evidence is examined, and

how researchers prefer to move between analytical results. Developing a separate application therefore offers an opportunity to investigate whether a more focused computational workflow can better support the tasks and preferences of local chemists.

This approach is related to *user-centered software design*, in which the needs, activities, and expectations of intended users guide system requirements and interface decisions. Rather than beginning with a list of software features and assuming they will be useful, the development process considers the questions users need to answer and the operations required to answer them. Research on end-user software engineering and interactive graph exploration provides a foundation for understanding how software functionality and interaction design influence users' ability to work with complex computational information [20, 29]. In this project, those principles are applied to chemists who need to investigate molecular features and spectral relationships without losing access to the underlying analytical evidence.

For example, a Wooster researcher investigating an unannotated molecular feature may want to locate that feature in a network, examine its precursor m/z and retention time, compare its MS/MS spectrum with those of neighboring features, and determine whether any nearby annotations provide useful chemical context. Each of these tasks requires connections between different kinds of information. A network visualization contains graph relationships, but the chemical evidence may be stored separately as spectra, feature tables, and annotation records. The proposed software must therefore coordinate these data sources so that researchers can move from a network-level overview to the evidence supporting an individual relationship.

This requirement introduces a software architecture problem. The application must define how scientific data are imported and validated, how spectral relationships are calculated or obtained, how graph structures are represented, and how results are presented to users. These responsibilities can be organized into separate software components. A data-processing component can manage imported analytical records and preserve their identifiers. A spectral-comparison component can evaluate fragmentation relationships using documented algorithms. A graph-processing component can store and analyze the resulting connections, potentially using a library such as NetworkX. Finally, a web interface can translate these computational results into interactive visualizations and chemical-information panels. Separating responsibilities can make the workflow easier to test, maintain, and modify without unnecessarily changing unrelated operations.

An additional design objective is to support scientific traceability. A researcher selecting a network edge should be able to investigate more than its displayed similarity value. The application should provide access to the associated features, supporting spectral information, and relevant computational settings. Likewise, users should be able to preserve an analysis with enough information to understand how it was generated and recover its results for future investigation. These capabilities connect software engineering decisions directly

to scientific interpretation, because the usefulness of a network depends on whether its underlying evidence remains available and understandable.

The motivation for a Wooster-specific system is therefore not to reproduce every capability of MS-DIAL or GNPS2. The goal is to develop a smaller, focused workflow that reflects how Wooster chemists investigate environmental HRMS data. Such a system can prioritize the operations most relevant to the laboratory, including feature-level exploration, transparent similarity calculations, accessible chemical metadata, persistent analyses, and predictable exports. Existing software may already perform individual parts of these tasks, but the research question concerns whether their integration into a locally designed application can provide a more consistent and useful experience for the intended researchers. Decisions about which operations to implement independently and which existing capabilities to reuse must be supported by technical evaluation.

The project also provides an opportunity to investigate how software usefulness can be evaluated in a scientific setting. Computational tests can examine whether feature identifiers, similarity scores, and graph structures are handled correctly. Repeated analyses can assess whether equivalent inputs and settings produce consistent results. Export and recovery tests can examine whether scientific information remains accessible after an analysis has been saved. In addition, representative chemical tasks and feedback from intended users can help evaluate whether the interface supports practical research questions. These assessments connect traditional software-engineering concerns with the needs of chemical investigation, rather than treating usability or reliability as secondary considerations.

Ultimately, the proposed Wooster molecular-networking application is intended to function as a research tool designed with chemists' analytical practices in mind. Its chemistry contribution lies in supporting the investigation and prioritization of known and unknown features in environmental HRMS datasets. Its computer science contribution lies in designing, implementing, and evaluating a computational workflow that connects data integrity, spectral comparison, graph analysis, interactive visualization, and reproducibility. The objective is not to claim that existing tools are scientifically inadequate, but to determine whether a locally tailored system can make molecular-networking analyses more accessible, transparent, and useful for the researchers who perform them.

1.6 Research Question and Specific Aims

The central research question guiding this interdisciplinary Senior Independent Study is:

> What computational workflow and interface are needed to make molecular networking reliable, reproducible, and useful for Wooster's environmental HRMS research?

1.6.1 Goal

The overarching goal of this project is to improve the interpretation of organic contaminants and other chemical features observed in surface-water samples from remote high-altitude environments by developing and evaluating a molecular-networking software workflow tailored to the College of Wooster's research needs. Mountain environments, including the Himalayas and the Qinghai–Tibetan Plateau, are relevant to studies of pollutant transport and deposition, although contamination at a particular site may reflect both atmospheric inputs and local activity [9, 41, 22]. In the available LC-HRMS datasets, suspect screening and spectral-library searches can support annotations of known chemicals, but many detected features remain without sufficiently supported structural assignments [23, 31, 17].

The working hypothesis is that a transparent, reproducible workflow combining existing feature annotations with appropriately filtered MS/MS spectral comparisons will reveal useful relationships among some annotated and unannotated features, helping chemists prioritize candidates for further investigation [36, 3, 8]. This hypothesis concerns the *usefulness of spectral relationships and the software that exposes them*, not a presumption that most unknown features can be structurally identified from a network. The research therefore has two linked outcomes: a tested computational tool for constructing and investigating molecular networks, and evidence-based chemical case studies using the high-altitude datasets.

For this study, reliability refers to the correct handling of analytical data, consistent execution of specified computational operations, and appropriate responses to incomplete or invalid inputs. Reproducibility refers to the ability to reconstruct analytical results using documented input data, software versions, and processing parameters. Usefulness concerns whether the workflow helps Wooster chemists investigate molecular features, examine spectral relationships, and access the evidence needed to interpret them. These properties provide a basis for evaluating the software independently of whether individual unknown features can be structurally identified.

The project combines analytical chemistry and computer science through five specific aims. The first two establish the experimental and chemical evidence available for investigation. The third and fourth focus on developing the computational workflow and interactive software, while the fifth evaluates the system's performance and its usefulness for environmental chemical research.

1.6.2 Aim 1: Characterize LC-HRMS/MS Data and Fragmentation Coverage

The first aim is to establish the analytical foundation required for molecular networking by characterizing the available liquid chromatography–high-resolution tandem mass spectrometry (LC-HRMS/MS) data from high-altitude surface-water samples. The research workflow uses reversed-phase C18 chromatography and

electrospray ionization (ESI) in positive and negative modes to provide complementary views of compounds with different chromatographic and ionization behavior. The experimental methods and origins of the existing samples and data will be distinguished from the computational work performed for this thesis. The relevant instrument settings, analytical procedures, and processing methods will be documented in Chapter 2 [12, 14, 26].

Where the existing acquisition records permit, the project will examine the fragmentation coverage provided by iterative data-dependent acquisition (DDA). In this strategy, successive runs can use inclusion and exclusion lists to direct MS/MS acquisition toward additional precursor ions rather than repeatedly fragmenting the same abundant signals. Automated exclusion-list methods described in the lipidomics literature provide a methodological precedent, although their performance in that application should not be assumed to transfer unchanged to these environmental water samples [21]. The analysis will consider which detected features have associated MS/MS spectra, which features lack sufficient fragmentation evidence, and how these differences influence their suitability for molecular networking.

Characterizing the available data also contributes to the computer science objectives of this project. Before implementing a molecular-networking workflow, it is necessary to understand how analytical features, fragmentation spectra, sample identifiers, and acquisition metadata are represented across the processed datasets. Different experimental runs may contain separate feature identifiers, while a single feature may have multiple associated spectra or no usable MS/MS spectrum. These relationships affect how the proposed software must organize, validate, and connect the imported information.

The expected outcome of this aim is a traceable description of the processed datasets, their fragmentation coverage, and their suitability for downstream spectral comparisons. This characterization will establish the analytical requirements for the software's data model and import procedures without claiming exhaustive chemical detection or identification [13, 31].

1.6.3 Aim 2: Annotate Known Contaminants Using Existing Reference Resources

The second aim is to establish a chemical-annotation baseline using suspect screening, existing spectral libraries, and the analytical evidence available for each feature. The upstream workflow uses MS-DIAL for feature detection, spectral processing, alignment, and related analytical operations before candidate annotations and associated spectra are examined for molecular-networking analysis [35, 33]. Suspect lists and reference resources can guide the investigation of environmental contaminants of concern, including

pharmaceuticals, pesticides, and industrial chemicals. However, a match to a database entry must be distinguished from a confirmed identification [23, 14, 26].

Candidate assignments will be evaluated using available evidence such as precursor accurate mass, isotope patterns, MS/MS fragmentation, retention behavior when informative, and comparison with appropriate reference spectra or standards [19, 17]. Identification confidence will be communicated using the Schymanski framework, which distinguishes confirmed structures from tentative structural assignments and less specific molecular information [32]. Features without sufficient evidence for a confident annotation will remain explicitly unannotated rather than being assigned unsupported chemical identities.

This aim also establishes how existing chemical information should be represented within the proposed software. Candidate names, reference-library information, precursor measurements, associated spectra, and identification confidence must remain connected to the correct analytical features. Preserving these relationships will allow the molecular-networking interface to distinguish annotated features from unannotated ones while retaining the evidence supporting each candidate assignment.

The expected outcome is a documented collection of annotated features and their associated confidence levels. These annotations will provide a chemical starting point for interpreting molecular networks and a baseline for evaluating whether spectral relationships offer additional information beyond direct library matches [6].

1.6.4 Aim 3: Develop a Molecular-Networking Workflow Beyond Direct Library Matches

The third aim is to design and implement the core computational workflow that compares MS/MS spectra and converts supported spectral relationships into a molecular network. Rather than relying exclusively on matches to known library entries, this approach investigates relationships among detected features, including those without confident annotations. Established molecular-networking methods demonstrate how spectral similarity can support the exploration of chemically related features and the prioritization of unknowns for further investigation [36, 27, 11].

A central challenge is developing a consistent computational representation of the analytical data. The software must accept defined outputs from the upstream processing workflow and correctly associate molecular features with their corresponding MS/MS spectra and chemical metadata. This requires establishing a data model that distinguishes analytical features, fragmentation spectra, annotations, and the relationships calculated between them. Input validation will be necessary to identify missing information, inconsistent feature identifiers, and other problems that could affect the interpretation of network results. Stable identifiers

and traceable source records will help ensure that the software does not associate spectra with incorrect features.

Once the analytical records have been organized, the workflow will apply documented spectral preprocessing and comparison procedures. These operations will include suitable peak filtering, fragment matching within specified mass tolerances, and cosine-based similarity calculations. The resulting scores will be evaluated using configurable criteria that determine whether a relationship should be retained as a network edge. Such decisions are important because spectral noise, peak alignment, similarity measures, and filtering thresholds can change the resulting network [28, 3, 8]. The selected methods and parameters will be documented so that the calculations can be examined and reproduced.

The retained relationships will then be represented using graph data structures. Each node will correspond to a clearly defined molecular feature, while edges will represent spectral relationships that satisfy the selected computational criteria. Edge attributes can preserve similarity scores and supporting information, allowing researchers to distinguish the existence of a graph connection from the chemical evidence behind it. General-purpose graph-processing tools such as NetworkX provide methods for storing weighted graphs, examining neighboring nodes, and identifying connected components [15, 25]. The suitability of these methods for the Wooster workflow will be evaluated as part of the software-development process.

Computational efficiency is another consideration because the number of possible spectral comparisons increases rapidly with dataset size. Comparing every unique pair of n spectra produces $n(n-1)/2$ candidate comparisons before filtering, creating quadratic growth in the number of comparisons. The implementation will therefore be evaluated with attention to processing time, data organization, and the practical size of networks generated from representative datasets. Performance improvements will be considered where measured limitations justify them, rather than assumed to be necessary before the workflow has been tested.

The intended outcome is a reproducible molecular-network construction process that allows existing chemical annotations to provide context for unannotated features. The software will preserve the parameters and evidence used to generate spectral relationships so that researchers can investigate why individual connections exist. Shared fragments and precursor-mass differences may motivate hypotheses about possible structural relationships, but the computational workflow will not treat these associations as proof of molecular identity or environmental transformation pathways [32, 6].

1.6.5 Aim 4: Build an Interactive, Chemist-Centered Software Interface

The fourth aim is to develop an interactive website that allows chemists to construct, inspect, interpret, and preserve molecular-networking analyses without losing access to the underlying experimental evidence. While

existing platforms such as MS-DIAL and GNPS2 provide substantial analytical and molecular-networking capabilities, the proposed application will focus on the particular tasks and preferences of researchers at the College of Wooster. The objective is not to reproduce the complete functionality of existing platforms, but to develop and evaluate a more focused workflow that supports how local chemists investigate environmental HRMS data.

This approach is informed by user-centered software design, in which the requirements of intended users guide decisions about functionality and interaction. Understanding the questions chemists want to answer is important for determining which software features are necessary and how those features should be organized. Research on end-user software engineering and interactive graph exploration provides a foundation for designing tools around the activities of domain specialists rather than requiring users to adapt their research practices to an unfamiliar computational interface [20, 29].

For example, a researcher investigating an unannotated feature may want to locate that feature within a molecular network, examine its precursor m/z and retention time, inspect neighboring features, and compare the fragmentation evidence supporting individual relationships. The interface must therefore connect the visual representation of the network to the analytical records from which it was constructed. Selecting a node should provide access to relevant feature information and available annotations, while selecting an edge should reveal the corresponding spectral relationship, similarity score, and supporting computational evidence.

Supporting these interactions requires a software architecture that coordinates several responsibilities. A data-management component must organize imported feature records, spectra, and metadata. A computational component must perform or retrieve the spectral comparisons and construct the resulting graph. Graph-processing functionality must support the analysis and retrieval of relationships, while the browser interface must present these results in an understandable form. Separating these responsibilities can help make the software easier to test, maintain, and extend as the research requirements develop.

The application will also distinguish between changes to the displayed network and changes to the underlying scientific analysis. For example, hiding selected edges in the interface may simplify visualization without changing their previously calculated similarity scores. In contrast, modifying spectral-matching parameters or network-construction thresholds may require generating a new analytical graph. Making these distinctions clear will help prevent users from confusing visualization choices with computational changes to the scientific results. Graph-layout methods will be selected to support readable exploration rather than treated as direct measurements of chemical similarity [18, 29].

Another primary objective is the preservation and recovery of analytical results. The proposed workflow will investigate how input data, network relationships, processing parameters, and associated chemical information can be saved and exported in a form that supports subsequent inspection. These capabilities are

important because a reproducible research tool must preserve more than the visual appearance of a network. It must also retain sufficient information to explain how the analysis was generated and allow its results to be reconstructed under documented conditions [40].

The expected outcome is a chemist-centered molecular-networking application that connects interactive visualization with analytical evidence, computational transparency, and reproducible data handling. Its design will prioritize the needs identified within the Wooster research workflow while retaining the flexibility to incorporate additional capabilities as those needs become better understood. The implemented interface and its effectiveness will be assessed through the software evaluation described in Aim 5.

1.6.6 Aim 5: Validate the Software and Apply It to Environmental Case Studies

The fifth aim is to evaluate whether the developed molecular-networking workflow is computationally reliable, reproducible, and useful for environmental chemical research. This evaluation will combine controlled software tests with representative analytical datasets and selected chemical case studies. The goal is to establish the strengths and limitations of the implemented system using documented evidence rather than assuming that its intended features will operate correctly or improve the research workflow.

Computational correctness will be evaluated through tests of data import, feature identification, spectral comparisons, and graph construction. Controlled examples with known expected results will be used to verify calculations, relationships, and the handling of incomplete or invalid information. Tests will examine whether feature identifiers remain associated with the correct MS/MS spectra, whether similarity calculations follow the selected algorithms, and whether network edges satisfy the documented filtering criteria. Graph-level comparisons may include node counts, edge sets, edge weights, isolated features, and connected components. Where appropriate, selected results from established molecular-networking software will provide a comparison baseline, with differences interpreted in relation to preprocessing and algorithmic settings.

Reproducibility will be evaluated by repeating analyses under equivalent input conditions and computational parameters. The resulting graph structures, similarity scores, metadata, and exported information will be compared to determine whether the software produces consistent analytical outputs. This evaluation will distinguish the underlying graph from its visual arrangement, since a graph-layout algorithm may place nodes differently without changing the scientifically relevant relationships. Tests of saving, reopening, and exporting analyses will also examine whether essential computational and chemical information can be recovered accurately [40].

Performance evaluation will consider how the workflow behaves as the amount of analytical data increases.

Measurements may include execution time, memory requirements, the number of processed spectra, and the size of the resulting networks. These observations will help determine whether the selected algorithms and data structures are suitable for the scale of the Wooster datasets. Research on similarity search and interactive graph exploration provides useful context for understanding computational and visualization constraints, although this project does not assume that every optimization described in the literature must be implemented [2, 29].

The usability evaluation will focus on whether the interface supports the analytical tasks for which it was designed. Representative activities may include locating an unannotated feature, examining its spectral neighbors, inspecting similarity evidence, reviewing candidate annotations, and recovering a previously saved analysis. Where feasible, feedback from Wooster researchers will be used to assess whether the application presents information clearly and supports their preferred workflow. This evaluation will consider scientific usefulness alongside software behavior, recognizing that an interface may perform its calculations correctly without necessarily meeting the needs of its intended users.

The chemistry evaluation will use selected case studies from the high-altitude environmental datasets. One case study will focus on features with existing candidate annotations, while another will examine features that remain without confident library matches. These investigations will evaluate whether molecular networking reveals useful spectral relationships, provides context for candidate structures, or helps prioritize features for additional analysis. The interpretation will distinguish computational associations from confirmed chemical identities and report uncertainty using appropriate analytical evidence and identification-confidence principles [13, 32, 6].

The expected outcome is a documented assessment of the proposed software's computational correctness, reliability, reproducibility, performance, and usability, together with evidence of its value and limitations in the selected environmental investigations. Success will not depend on identifying every unknown compound or demonstrating that the new website is universally superior to existing platforms. Instead, the evaluation will determine whether a Wooster-specific molecular-networking workflow can provide dependable computational results and meaningful support for the chemical questions motivating this Senior Independent Study.

1.7 Project Scope and Thesis Contributions

1.7.1 Project Scope

This Senior Independent Study focuses on developing and evaluating a computational workflow for molecular networking using existing environmental LC-HRMS/MS data. The project includes examining the

requirements of Wooster chemists, defining and validating data inputs, implementing spectral comparison and network-construction procedures, and developing an interactive website for investigating molecular features and their relationships. Selected environmental datasets will provide the basis for evaluating both the computational workflow and its usefulness for chemical interpretation.

The proposed software is intended to complement established analytical tools rather than replace their complete functionality. MS-DIAL will continue to provide upstream mass spectrometry data processing, while existing molecular-networking platforms such as GNPS and GNPS2 provide important scientific methods and potential comparison baselines. The Wooster application will focus on the computational operations and user interactions most relevant to local researchers, including access to spectral evidence, preservation of feature metadata, reproducible network construction, and reliable saving and exporting of analyses. The specific software components, input formats, algorithms, and evaluation procedures will be described in Chapter 2.

The project also has important scientific boundaries. Molecular networking cannot independently confirm the identity of an unknown compound, establish a chemical transformation pathway, or determine the environmental source of a detected feature. Likewise, software development does not establish chemical toxicity, ecological effects, or exposure levels. The chemical case studies will therefore distinguish experimental observations, candidate annotations, and computationally suggested relationships from conclusions supported by confirmatory chemical evidence.

1.7.2 Expected Computer Science Contributions

The primary Computer Science contribution is the design, implementation, and evaluation of a molecular-networking workflow and interactive website tailored to the research needs of College of Wooster chemists. This work involves translating scientific requirements into a software system that connects analytical data processing, spectral similarity calculations, graph representation, and interactive visualization. Particular attention will be given to the design of data structures, reliable associations between features and spectra, documented algorithms, and the organization of computational components.

The research will also examine software reliability, reproducibility, and usability. Input validation and controlled algorithmic tests will assess computational correctness, while repeated analyses and export procedures will evaluate whether results and their supporting information can be preserved and reconstructed. User-centered design principles will guide the development of interactions that allow chemists to investigate network relationships and access relevant experimental evidence. Together, these efforts aim to establish a transparent and maintainable computational workflow rather than simply produce another molecular-network visualization.

The significance of the Computer Science contribution will be evaluated through the implemented functionality, computational tests, performance measurements, and evidence of how effectively the software supports the intended research tasks. The resulting findings will also help identify the strengths, limitations, and potential future improvements of a locally developed scientific application.

1.7.3 Expected Chemistry Contributions

The primary Chemistry contribution is the application of molecular networking to existing environmental LC-HRMS/MS datasets from remote and high-altitude surface-water environments. Building on previous analytical research, the project will examine spectral relationships among features with existing candidate annotations and features that remain unannotated after reference-based screening. These investigations may provide additional chemical context for known contaminants and help prioritize unknown features for further study.

Selected case studies will examine the available analytical evidence, including precursor masses, MS/MS fragmentation patterns, candidate annotations, and relationships within molecular networks. The analysis will consider whether connected features share informative spectral characteristics and whether these relationships support plausible hypotheses about chemical similarity. Identification confidence will be communicated using the Schymanski framework, and computational relationships will not be presented as confirmed molecular structures without appropriate evidence.

The intended contribution is therefore not a comprehensive identification of every chemical detected in the environmental samples. Instead, the project aims to demonstrate how transparent molecular-networking methods can complement existing suspect-screening approaches, support chemical interpretation, and guide further investigation of complex environmental datasets.

1.7.4 Organization of the Thesis

The remainder of this thesis is organized into four chapters. Chapter 2 describes the research design, environmental datasets, analytical methods, software requirements, system architecture, spectral comparison algorithms, graph construction, and evaluation procedures. Chapter 3 presents the computational implementation and experimental results, including software validation, performance measurements, molecular-network characteristics, and selected environmental chemical case studies. Chapter 4 discusses the findings from both disciplines, evaluates the software's design decisions and limitations, and considers the implications for chemical interpretation and future computational research. Finally, Chapter 5 summarizes the principal

contributions, addresses the central research question, and identifies opportunities for further development of the Wooster molecular-networking workflow.

CHAPTER 2

Methodology, Research Design, and Software Development

2.1 Overall Research Design and Workflow

2.2 Environmental Samples and Chemistry Experimental Methods

2.3 Upstream Data Processing and Chemical Annotation

2.4 Software Requirements and System Architecture

2.5 Spectral Comparison and Similarity Calculation

2.6 Molecular-Network Construction

2.7 User Interface Design and Interaction

2.8 Software Implementation and Reproducibility

2.9 Software Evaluation and Testing Methods

2.10 Chemical Case-Study Design

CHAPTER 3

Results and Data Analysis

3.1 Environmental Dataset and Feature Summary

3.2 Software Implementation Results

3.3 Computational Correctness and Validation

3.4 Software Performance and Workflow Evaluation

3.5 Molecular-Network Structure and Characteristics

3.6 Chemical Case Study 1: Annotated Molecular Features

3.7 Chemical Case Study 2: Unidentified Molecular Features

3.8 Summary of Principal Results

CHAPTER 4

Discussion

4.1 Interpretation and Integration of Results

4.2 Computational Contributions and Design Trade-Offs

4.3 Chemical Significance and Comparison with Literature

4.4 Study and Implementation Limitations

4.5 Implications for Future Environmental HRMS Research

CHAPTER 5

Conclusions and Future Work

5.1 Summary of the Research

5.2 Principal Computer Science Contributions

5.3 Principal Chemistry Contributions

5.4 Answer to the Central Research Question

5.5 Future Work

5.6 Final Remarks

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